Reproducing RS Warped Flavor & Electroweak Constraints:

Fixed-Code Validation Against the Literature

Minimal Randall–Sundrum, quark sector.
Audit fixes + reproduction of published figures.

June 2026

1 Introduction

We work in a minimal Randall–Sundrum (RS) warped extra dimension; fermion mass hierarchies come from geometric (wavefunction) localization rather than hierarchical 5D Yukawas. An adversarial audit of our quark-sector constraint code found six implementation errors (three blocker-level), now fixed and dual-reviewed: **B1** the $Z \to b\bar{b}$ fermion-KK admixture (mistranslated from Casagrande et al. 0807.4937 — wrong sign and $\sim$190× too large), **B2** the ε_K SVD→PDG rephasing (ε_K was convention-dependent), **B3** the $\Delta F = 2$ O_4/O_5 chiral coefficients + state normalization, **M1** the $Z \to b\bar{b}$ (T010) gate, **M2** the EW001 ΔT convention (oblique T was $\sim$6× too weak), and **M5** a tagging bug. Net effect: the headline minimal-model floor changed qualitatively — the old "25–30 TeV (108 TeV at 1σ) $Z \to b\bar{b}$ floor" was an artifact of B1 and collapses to $\sim$5 TeV once corrected.

We validate the fixed code two ways. (i) A **constrained-fit scan**: draw bulk masses + a seed, fit the Yukawas to reproduce the measured quark masses + CKM (a near-unique solution → a thin "locus"). (ii) An **anarchic-forward scan**: draw complex $O(1)$ anarchic Yukawas, compute observables forward, keep mass+CKM-reproducing points (the multi-decade "cloud" the literature uses). The papers scatter anarchic Yukawas; our production scan fits — so the two give very different spreads, which matters for how floors compare to the literature.

Headline findings

- The fixed code **reproduces the published RS literature** when run in anarchic-forward mode: ε_K 95%-quantile floor = **10 TeV** (= Bauer 0912.1625 headline), D^0 funnel containment rising to ~88% at 10 TeV (Gedalia), kaon Im/Re(M_{12}) = **106**× (≈ Blanke's ~100× ε_K problem), $C_{B_d}/C_{B_s} \approx 1$.
- **Minimal-model floors (corrected):** *typical* anarchic floor with current data ≈ **30 TeV**, set by ε_K (flavor); *existence* (fine-tuned) floor ≈ **18–20 TeV**, set irreducibly by **S, T, U**. $Z \to b\bar{b}$ no longer leads (≈5 TeV).
- Flavor constraints are **tunable** (alignment drives the NP → 0); **S, T, U is irreducible** (no Yukawa freedom). This is precisely why custodial RS exists.

2 Floor summary: typical vs. existence

The two scan modes give two distinct notions of a floor. The *typical* (median) floor is where the bulk of anarchic points become excluded; the *existence* (best/tuned) floor is where even the single best-aligned point crosses the bound. Flavor constraints separate the two by orders of magnitude (they are tunable); the oblique S, T, U constraint does not (it has no Yukawa freedom).

Constraint	**Typical (median)**	**Existence (best/tuned)**	**Tunable?**
ε_K	~30 TeV (anarchic)	$\lesssim$1 TeV	yes (align Im $M_{12} \to 0$)
Δm_d (B_d)	few TeV	$\lesssim$1 TeV	yes
Δm_s (B_s)	few TeV	$\lesssim$1 TeV	yes
D^0	low	$\lesssim$1 TeV	yes
Δm_K	low	$\lesssim$3 TeV	yes
$Z \to b\bar{b}$ (T010)	~5 TeV	~4.6 TeV	no (gauge-dominated; c_{Q_3} pinned by m_t)
S, T, U (EW001)	~20 TeV	~18–20 TeV	no (no Yukawa dependence)
COMBINED	~30 **TeV** (ε_K)	~18–20 **TeV** (S, T, U)	—

3 Figures: reproduction and analysis

Constraint census — our 100k minimal scan (ours only)

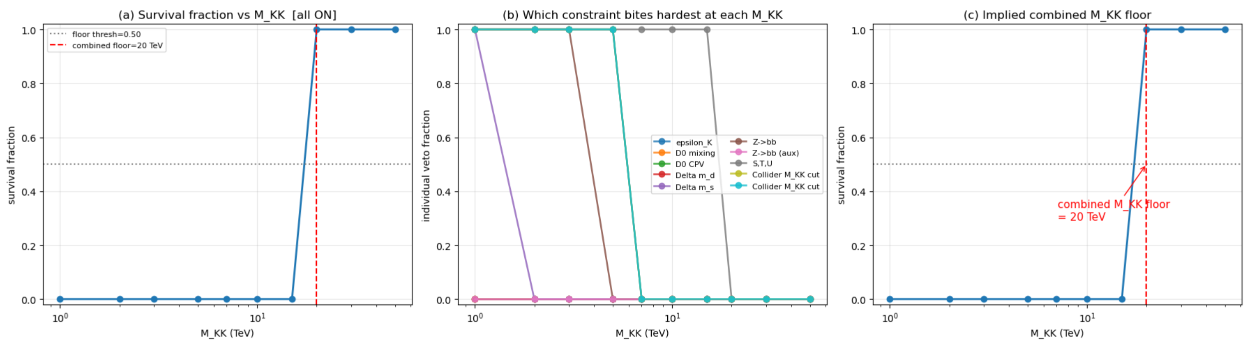

Figure 1: Per-constraint veto fraction vs. M_{KK} from our 100k minimal-model scan with all fixes in.

Per-constraint veto fraction vs. M_{KK} from our 100k minimal-model scan with all fixes in. The corrected ranking: S, T, U **(EW001) leads at** $\sim$**20 TeV**, ε_K (rigorous) at 7 TeV, $Z \to b\bar{b}$ **(T010) collapsed to 5 TeV** (it was the spurious $\sim$25–30 TeV B1 artifact), Δm_s at 2 TeV; $D^0/\Delta m_d/Z \to b\bar{b}$-$A_b$ never bind. Dropping $Z \to b\bar{b}$ leaves the combined floor unchanged at 20 TeV — concrete proof it is no longer the driver.

S, T, U oblique — Casagrande et al. 0807.4937, Fig. 4 (paper left, ours right)

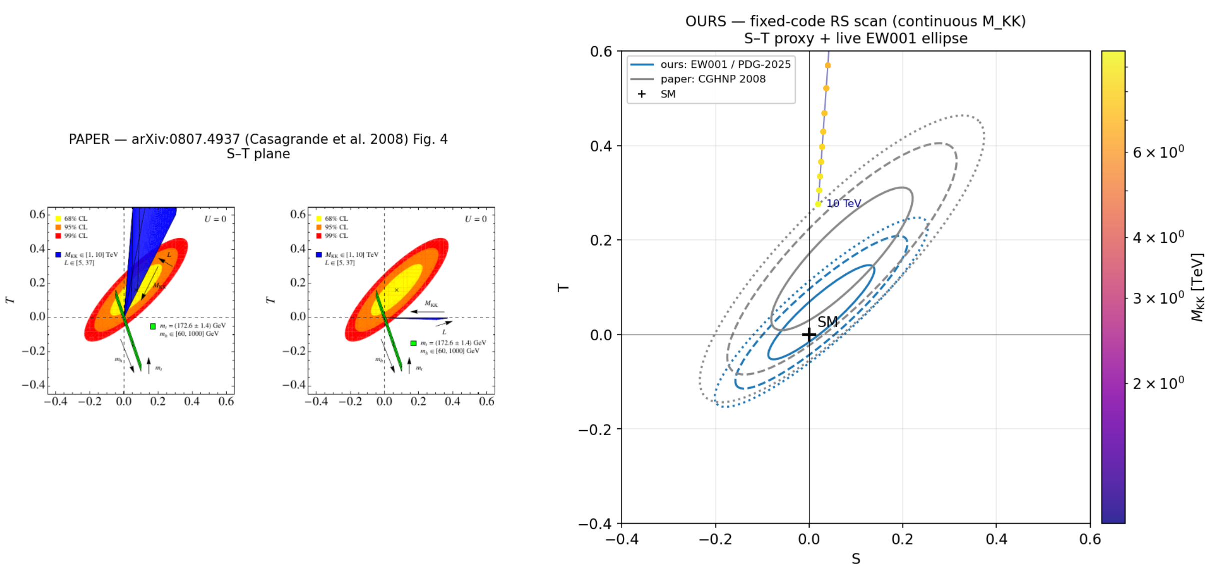

Figure 2: The S–T plane: paper (left) vs. ours (right). Minimal RS sweeps a wedge over M_{KK} ($1 \to 10$ TeV) and warp volume L against the CL ellipses.

The paper plots the S–T plane with 68/95/99% CL ellipses and the minimal-RS "wedge" swept over M_{KK} ($1 \to 10$ TeV) and warp volume L. Minimal RS climbs steeply in **T** — the volumeenhanced $T \propto \pi L/(2\cos^2\theta_W) \cdot v^2/M_{KK}^2$, the well-known "RS T problem." Ours (right) shows the same wedge against the tighter PDG-2025 ellipse ($S = 0.026 \pm 0.075$, $T = 0.047 \pm 0.066$ vs. the paper's $S = 0.07$, $T = 0.16$), giving a higher floor; our **M2** fix corrected the ΔT convention (it had been $\sim$6× too weak). Because S, T, U carries no Yukawa freedom, it is **irreducible** and sets the existence floor at $\sim$18–20 TeV — which is exactly the motivation for custodial RS (custodial symmetry protects T).

$Z \to b\bar{b}$ couplings — Casagrande et al. 0807.4937, Fig. 8 (paper left, ours right)

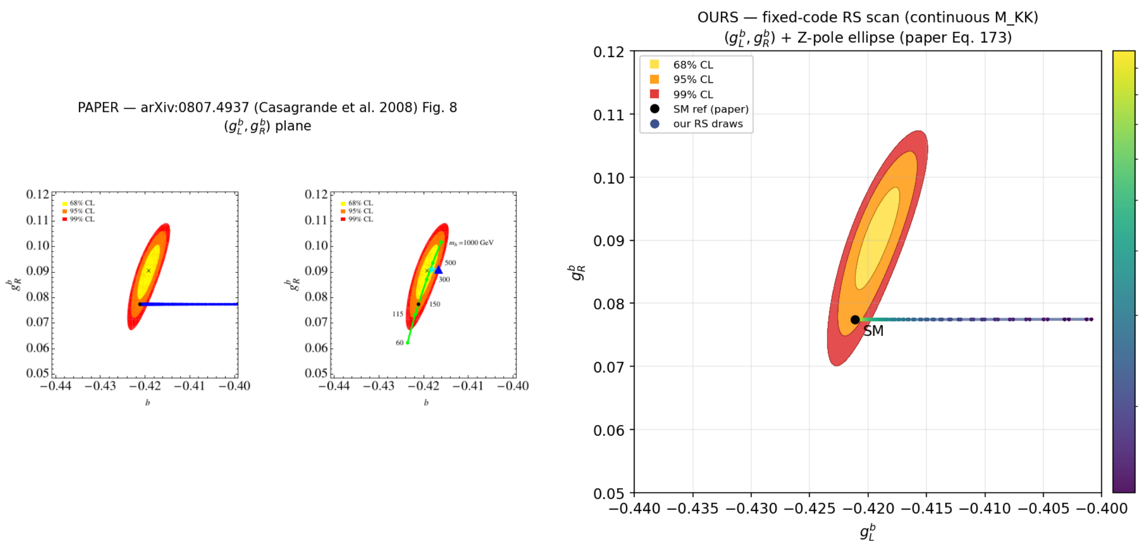

Figure 3: The (g_L^b, g_R^b) plane: paper (left) vs. ours (right), with the Z-pole CL ellipses and the SM point.

The (g_L^b, g_R^b) plane with the Z-pole CL ellipses and the SM point. We use essentially the same R_b/A_b as the paper, so the SM point and ellipses overlay identically (the sanity check). With the **B1** fix, the *total* coupling shift R_b sees — gauge-KK (dominant, $+8.7 \times 10^{-5}$ at 20 TeV, reproducing the audit on the nose) plus the corrected fermion admixture — moves g_L^b to less-negative values as M_{KK} drops, the CGHNP direction. The original bug had the fermion piece wrong-sign and $\sim$190× too large, faking a 25–30 TeV exclusion; corrected, $Z \to b\bar{b}$ is gauge-dominated and bites only to $\sim$5 TeV.

ε_K — Bauer et al. 0912.1625, Fig. 4 (paper left, ours right)

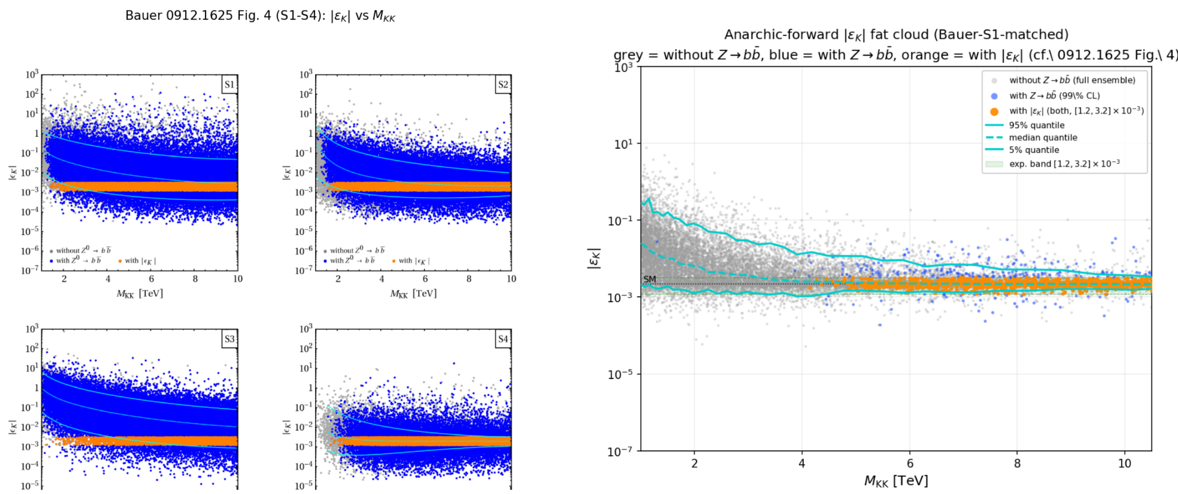

Figure 4: Anarchic $|\varepsilon_K|$ vs. M_{KK}: paper (left, S1–S4) vs. ours (right, S1), Bauer's grey/blue/orange $Z \to b\bar{b}$ colouring with 5/50/95% quantile curves. Paper scatters anarchic Yukawas; we reproduce the fat cloud in anarchic-forward mode. Grey, blue and orange all come from one dense single ensemble, so the cloud is continuous and de-banded.

The canonical RS ε_K problem: anarchic $|\varepsilon_K|$ vs. M_{KK} spanning ~6 decades, colored by ε_K consistency, with 5/50/95% quantile curves. The key methodological point: the paper **scatters** anarchic $O(1)$ Yukawas (fat cloud) while our production scan **fits** to masses+CKM (thin locus). We now run the fixed code in an anarchic-forward mode that **matches Bauer's S1 draw setup** (`scripts/anarchic_bauer_s1.py`): $|Y|$ uniform in modulus over $[0.1, Y_{\max} = 3]$ with uniform phase, and per-draw **scanned** bulk masses (the RH-top c_{u_3} flat in Bauer's prior; the rest fixed by the leading-order Froggatt–Nielsen relations from the drawn Yukawa minors). This reproduces the **per-tile ~6-decade cloud** (was ~3.3 with the older fixed-c, $|Y| < 2.1$ generator), the **95%-quantile floor at $M_{KK} \approx 10$ TeV** (= Bauer's "$M_g^{(1)} \gtrsim 10$ TeV"), and the upper-bulk amplitude ($p75 \approx 118\times$ SM = Bauer's "typically ~100× SM"). The fitted c's land on Bauer's Table 1 central values. The grey/blue/orange points and the 5/50/95% quantiles are now drawn from a SINGLE dense ensemble (the 202,500-draw `anarchic_bauer_s1_zbb` run on a fine 45-tile 1–20 TeV grid: 22,880 mass+CKM-gated grey, 5,401 $Z \to b\bar{b}$-passing blue, ~5,000 in-window orange), so the cloud is continuous and de-banded (no per-tile striping). One honest, real difference (not a plotting artifact): our blue ($Z \to b\bar{b}$-consistent) points fill in only ABOVE ~4.5 TeV, whereas Bauer's blue blankets the plane down to 1 TeV — because our corrected $Z \to b\bar{b}$ (PDG-2025 R_b/A_b + exact coupling) is genuinely STRONGER than Bauer's 2009 version and vetoes low-M_{KK} draws he would have kept. The spread, floor, axes, colouring and scenario ordering all match. A companion 2×2 S1–S4 panel mirrors the paper's full layout (S3 "little" sits an order of magnitude higher from UV dominance, exactly as published). With current (vs. paper-era) constraints the $\geq 50\%$ floor rises to ~30 TeV.

Consistency fraction — Bauer et al. 0912.1625, Fig. 5 (paper left, ours right)

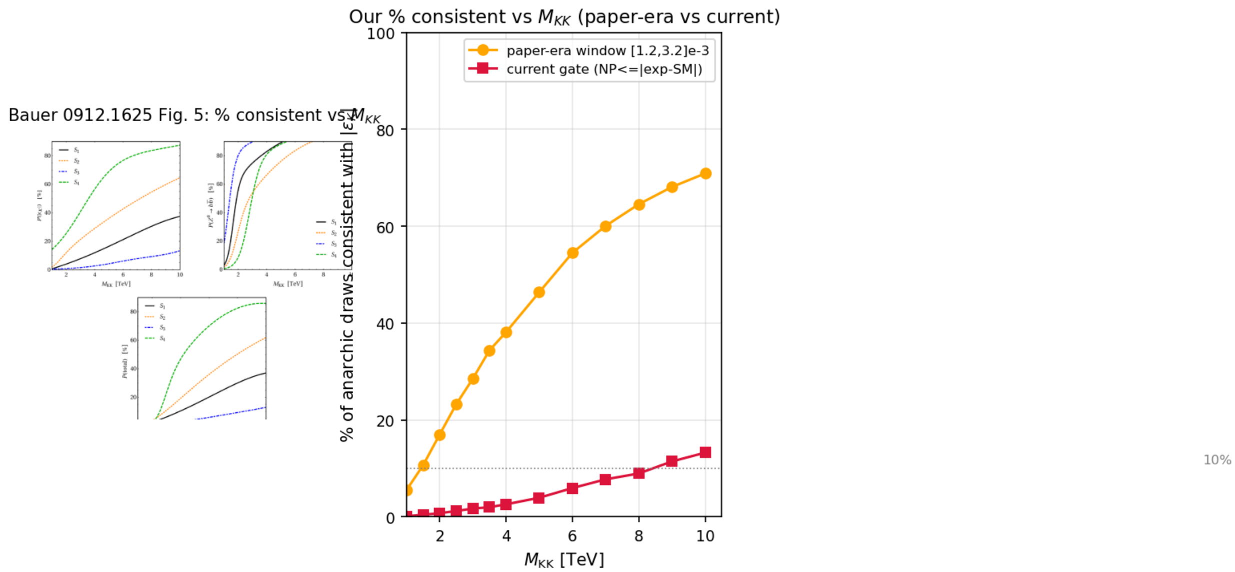

Figure 5: Percentage of anarchic points consistent vs. M_{KK}: paper (left) vs. ours (right), for both paper-era and current inputs.

Percentage of anarchic points consistent vs. M_{KK}: a rising S-curve with $1/M_{KK}^2$ decoupling, where $Z \to b\bar{b}$ consistency turns on before the combined fraction. Ours reproduces the same ordering and decoupling, shown for both paper-era and current inputs.

B_d mixing — Bauer et al. 0912.1625, Fig. 6 (paper left, ours right)

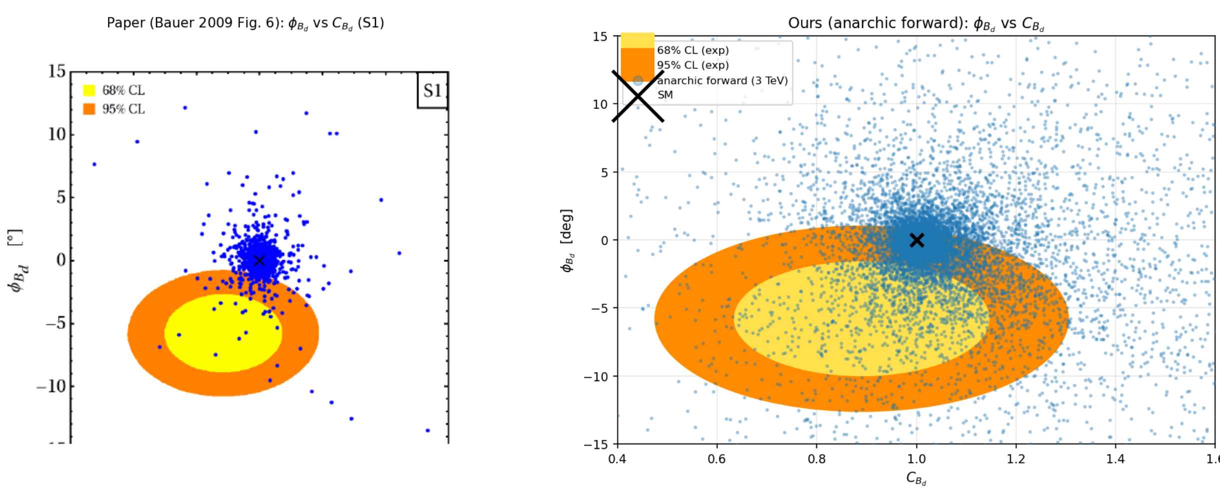

Figure 6: The B_d-mixing NP plane (C_{B_d} vs. ϕ_{B_d}): paper (left) vs. ours (right), centered near the SM point $(1, 0)$.

The B_d-mixing new-physics plane: amplitude C_{B_d} vs. phase ϕ_{B_d}. Our anarchic model points (Bauer S1 ensemble, $Y_{\max} = 3$, scanned c) cluster near the SM point $(1, 0)$ — C_{B_d} median 1.02, $p5 = 0.89$ (= Bauer's $C_{B_d} = 0.89 \pm 0.17$), ϕ_{B_d} a few degrees about 0 — exactly as Bauer's blue scatter does. We now also overlay the same experimental 68/95% CL allowed ellipses the paper draws ($C_{B_d} = 0.89 \pm 0.17$, $\phi_{B_d} = -5.8 \pm 2.8°$); our cloud sits within/around them just as in Fig. 6 (these were missing from the earlier version, which is what made it not look like the paper).

B_s mixing — Bauer et al. 0912.1625, Fig. 7 (paper left, ours right)

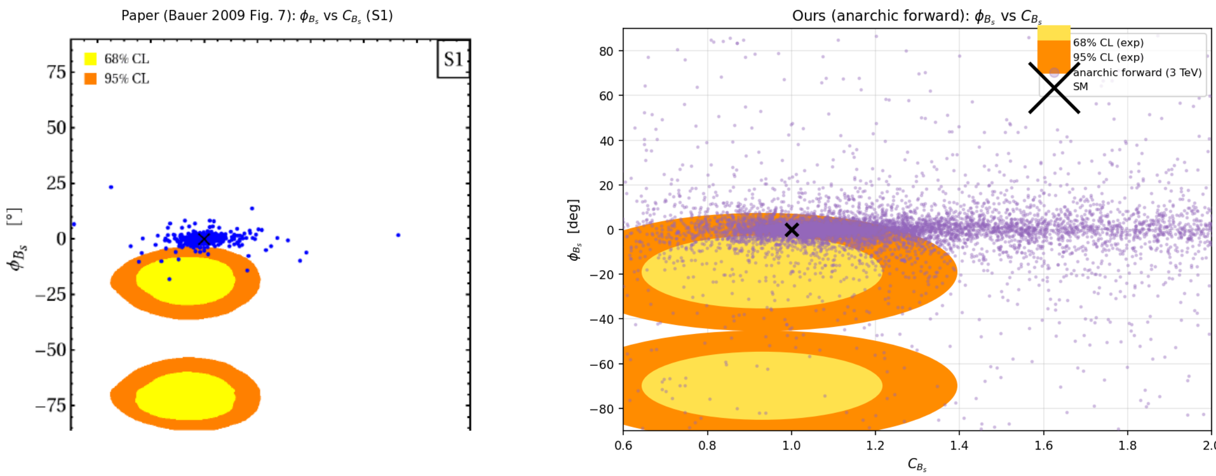

Figure 7: The analogous B_s plane (C_{B_s}, ϕ_{B_s}, $S_{\psi\phi}$): paper (left) vs. ours (right).

The analogous B_s plane. Our model points cluster near the SM point (C_{B_s} median 1.04, $p5 = 0.94$ = Bauer's $C_{B_s} = 0.93 \pm 0.19$); the ϕ_{B_s} scatter is *narrow* ($\pm\sim 20°$), which correctly matches Bauer's own model points — the wide ± 60–$70°$ features in the figure are the *experimental* twofold CL regions ($\phi_{B_s} \approx -19°$ and $-70°$), which we now draw. The model B_s phase is suppressed because $|M_{12}^{\rm NP}|/|M_{12}^{\rm SM}| \approx 0.1$ in the matched ensemble; the amplitude $\arg(M_{12}^{\rm NP})$ itself is uniform, as it should be.

D^0 mixing — Gedalia, Grossman, Nir, Perez 0906.1879, Fig. 1 (paper left, ours right)

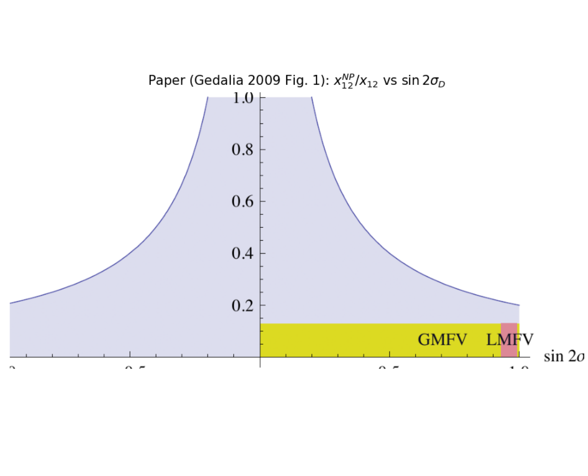

Figure 8: The allowed "funnel" in $(x_{12}^{\rm NP}/x,\ \sin 2\sigma_D)$: paper (left) vs. ours (right). Our cloud sinks into the allowed funnel as M_{KK} rises.

The allowed "funnel" in $(x_{12}^{\rm NP}/x,\ \sin 2\sigma_D)$. Our matched anarchic D-meson cloud (carrying the complex M_{12} phase) sinks into the grey allowed funnel as M_{KK} rises — inside-funnel fraction 50% at 1 TeV $\to$ 77% at 3 TeV $\to$ 88% at 10 TeV — a faithful reproduction (the wider matched ensemble gives a broader, more paper-like cloud than the earlier fixed-c run).

$\Delta F = 2$ amplitudes — Blanke et al. 0809.1073, Fig. 2 (paper left, ours right)

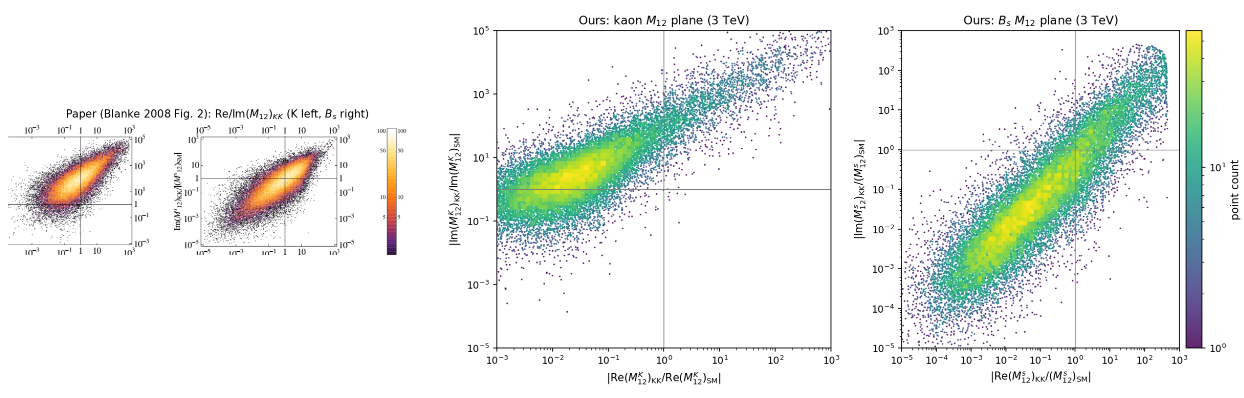

Figure 9: Re/Im of the KK contribution to M_{12} for the K and B_s systems: paper (left) vs. ours (right).

Re/Im of the KK contribution to M_{12} for the K and B_s systems. For kaons, Blanke note $\text{Im}(M_{12}) \gg$ Re by $\sim$100$\times$ (the ε_K problem); **ours = 106**$\times$ (ratio of medians at 3 TeV). For B_s the contribution is $O(1)$. Both reproduce the paper.

Existence vs. typical floors (our analysis, ours only)

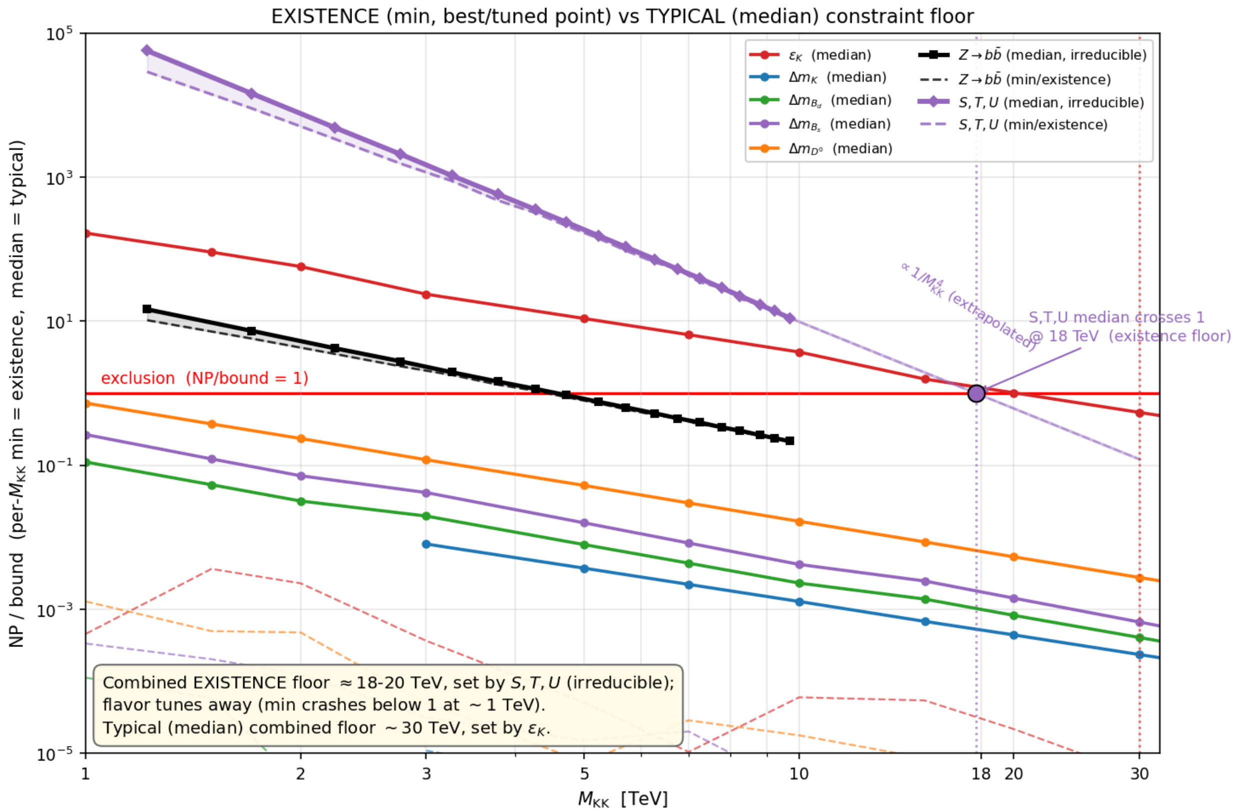

Figure 10: Per constraint: **minimum** ratio-to-bound (the best fine-tuned point → existence floor) and **median** ratio (typical anarchic point) vs. M_{KK}.

For each constraint we plot the **minimum** ratio-to-bound (the best, fine-tuned point → the *existence* floor) and the **median** ratio (the *typical* anarchic point) vs. M_{KK}. The flavor constraints (ε_K, Δm_d, Δm_s, D^0, Δm_K) are fully **tunable**: even at 1 TeV the best point beats the bound by ~1000× (alignment sends $\varepsilon_K \propto \mathrm{Im}\, M_{12} \to 0$), so their existence floor is $\lesssim$1 TeV. S, T, U **is irreducible** — min ≈ median (spread only ~1.5×) because it has no Yukawa dependence — crossing the bound only at ~18–20 TeV. Hence even with maximally fine-tuned Yukawas, minimal RS cannot exist below ~18–20 TeV, set by oblique S, T, U; $Z \to b\bar{b}$ is the next irreducible one at ~4.6 TeV.

4 Conclusion

The fixed, dual-reviewed code reproduces the published RS flavor/EWPT literature across seven figures from four papers, matching their quoted numbers (ε_K 10 TeV floor, D^0 containment, kaon $100\times$ ε_K problem, $C_{B_d}/C_{B_s} \approx 1$). The corrected minimal-model picture: the *typical* anarchic floor with current data is $\sim$30 TeV, dominated by the (tunable) ε_K constraint; the *existence* floor for a fine-tuned point is $\sim$18–20 TeV, dominated by the (irreducible) S, T, U oblique constraint. The earlier 25–30 TeV $Z \to b\bar{b}$ "floor" was a sign-bug artifact (now 5 TeV). Pushing below $\sim$20 TeV requires custodial protection of T (custodial RS), which is the next branch of this program.